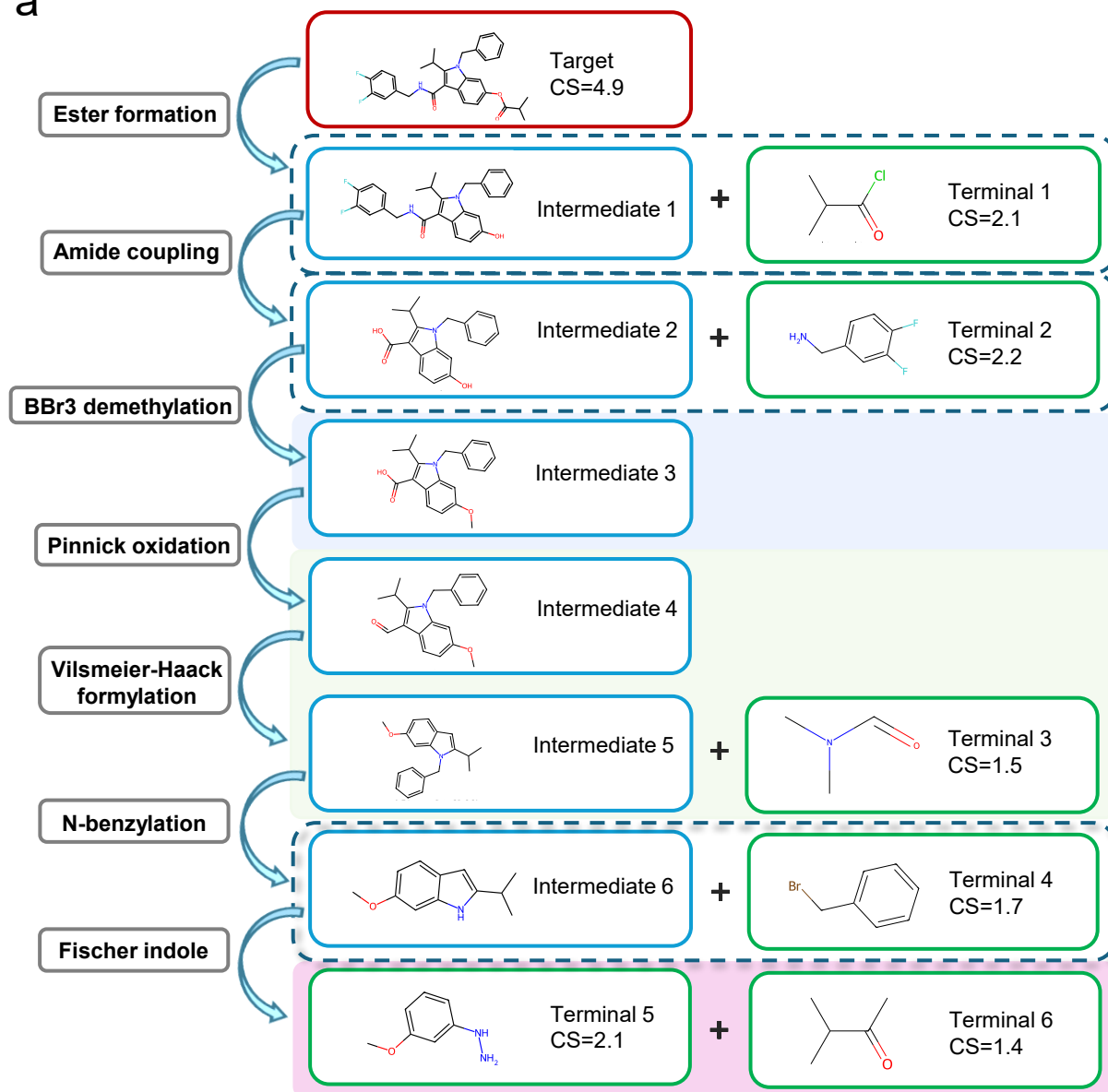


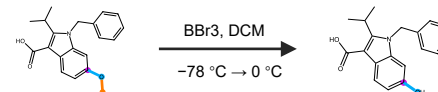
a



b

1 Masked phenol strategy: BBr₃ demethylation

LLM Rationale: Trace the C6 phenol back to a methyl ether. Retain OMe during upstream N-benylation and C3 functionalization to limit competing phenol reactivity.

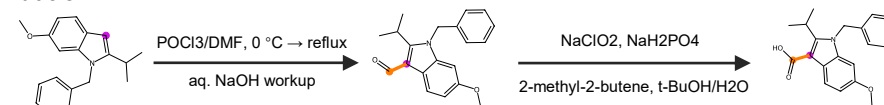


Route Implication: Unmask the phenol after C3 oxidation and before amide coupling and final O-acylation.

Alternatives Considered: TBS protection and an earlier retrosynthetic oxidation-state change were considered; OMe masking was selected to address phenol reactivity.

2 C3 carboxylic acid handle installation: Vilsmeier-Haack formylation → Pinnick oxidation

LLM Rationale: Trace C3-CO₂H back to C3-CHO, then to the unfunctionalized C3 position. Forward Vilsmeier-Haack formylation uses DMF as the carbon source, followed by Pinnick oxidation.

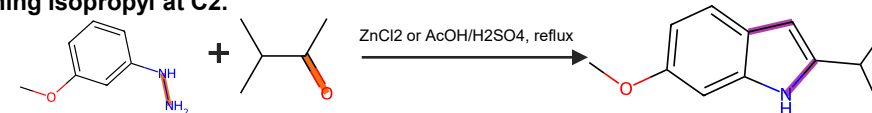


Route Implication: Install the carboxyl group required for amide coupling while retaining the OMe and N-benzyl groups.

Alternatives Considered: Duff and Reimer-Tiemann formylation were considered. A methyl-ester hydrolysis route was also considered but would require prior ester installation.

3 Indole core construction: Fischer indole synthesis

LLM Rationale: Disconnect the indole into 3-methoxyphenylhydrazine and 3-methylbutan-2-one. The depicted Fischer pathway maps OMe to C6 and the ketone's methyl-side α-carbon to C3, retaining isopropyl at C2.



Route Implication: Provide the indole core for subsequent N-benylation and C3 formylation; aryl cyclization regioselectivity remains to be established.

Alternatives Considered: The original record considered Madelung and Bischler-Möhlau routes and selected Fischer cyclization for the core-forming step.